

		6 cm produces resonance.
15	C	Since the orders overlap at the same angle: $n_1\lambda_1 = n_2\lambda_2$ $\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{400}{600} = \frac{2}{3}$ Therefore the second order 600 nm is overlapping with the third order 400 nm. $\text{Using } d = \frac{n\lambda}{\sin \theta} = \frac{2(600 \times 10^{-9})}{\sin 30^\circ} = 2.4 \times 10^{-6} \text{ m}$
16	B	Initial state.